Chapter 17: Physics of Hearing

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 17

Part I

Sound

Outline of Part I: Sound

What is Sound?

Speed of Sound, Frequency, and Wavelength

Outline of Part I: Sound

What is Sound?

Speed of Sound, Frequency, and Wavelength

Sound vs. Hearing

A Famous Thought Experiment

"If a tree falls in the forest and no one is around to hear it, does it make a sound?"

Sound (Physics)

- A disturbance of matter transmitted outward from its source.
- A physical phenomenon. It *exists* regardless of whether anyone hears it.

Hearing (Perception)

- The *sensory perception* of sound by an observer.
- Hearing is **observer-dependent**.
- Sound is not.

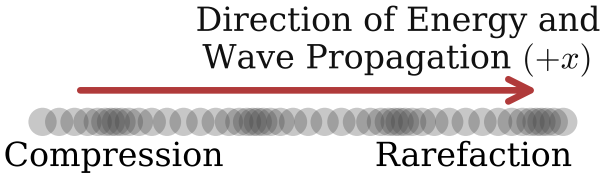
Sound is a Longitudinal Wave

Sound is a disturbance of matter that propagates outward from a vibrating source as a **longitudinal mechanical wave**.

- **Longitudinal**: particle oscillation is *parallel* to the direction of wave propagation.
- **Mechanical**: requires a medium (matter) to travel through.
- There is no sound in a vacuum.

Compression → high pressure

Rarefaction → low pressure

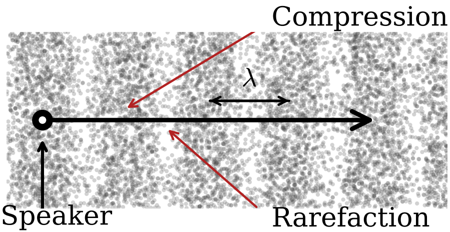


(Particles oscillate back and forth)

How Sound is Produced

Vibrating Source Creates Pressure Waves

- Source moves **forward**: atoms pushed together ⇒ **compression** (high pressure).
- Source moves **backward**: atoms spread apart ⇒ **rarefaction** (low pressure).
- These alternating pressure regions propagate outward at wave speed v_w.
- The wave frequency *equals* the source frequency.



Sound amplitude decreases with distance as energy spreads over a larger area and is partially absorbed by the medium.

Sound Requires a Medium

Sound requires matter to travel through.

There is no sound in a vacuum.

- Sound travels as a longitudinal wave and needs atoms to compress and rarefy.
- Different media have different wave speeds (see next slide).
- Generally: $v_{\text{solids}} > v_{\text{liquids}} > v_{\text{gases}}$

High-intensity sound can shatter glass at its resonant frequency — proof that sound, though invisible, exerts real physical force on matter.

Check Your Understanding

In a longitudinal sound wave, in which direction do air particles oscillate relative to the direction the wave travels?

How does this differ from a transverse wave?

Come to lecture to see the answer.

Outline of Part I: Sound

What is Sound?

Speed of Sound, Frequency, and Wavelength

Wave Speed, Frequency, and Wavelength

Wave Equation — SI Units: m/s, Hz, m

The wave speed v_w, frequency f, and wavelength λ are related by:

$$v_w = f\lambda$$

where v_w is the wave speed (m/s), f is the frequency (Hz), and λ is the wavelength (m).

- **Frequency** is set by the *source*; it does not change when sound enters a new medium.
- **Wavelength** adjusts when speed changes: $\lambda = v_w/f$.
- **Speed** depends on the medium.

Rearranging:

$$\lambda = \frac{v_w}{f} \qquad f = \frac{v_w}{\lambda}$$

Higher frequency $\Rightarrow$ shorter wavelength (for same v_w).
Lower frequency $\Rightarrow$ longer wavelength.

Speed of Sound Depends on the Medium

Factors Affecting the Speed of Sound

- **Rigidity**: more rigid media transmit compressions faster $\Rightarrow$ higher v_w.
- **Density**: denser media have more inertia per atom $\Rightarrow$ generally lower v_w.
- $\therefore$ solids are fastest; gases are slowest.

Approximate Wave Speeds

Medium	**Speed (m/s)**	**State**
Air (0°C)	331	Gas
Air (20°C)	343	Gas
Fresh water (20°C)	1480	Liquid
Seawater (25°C)	1540	Liquid
Steel	5960	Solid

Speed of Sound in Air

Speed of Sound in Air as a Function of Temperature

The speed of sound in air depends on temperature (T in kelvin):

$$v_w = \left(331\ \frac{\mathrm{m}}{\mathrm{s}}\right)\sqrt{\frac{T}{273\,\mathrm{K}}}$$

where $T = T(^\circ\mathrm{C}) + 273\,\mathrm{K}$ is the absolute temperature.

Temperature Conversion

$$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273$$

- At 0°C: $v_w = 331\,\mathrm{m/s}$
- At 20°C: $v_w \approx 343\,\mathrm{m/s}$

Why does temperature matter?

Higher temperature $\Rightarrow$ faster-moving molecules $\Rightarrow$ compressions transmitted more quickly $\Rightarrow$ higher wave speed.

Example: Wavelength of a 440 Hz Sound Wave

Find the wavelength of a 440 Hz sound wave in 15°C air. (cp2e_ch17_ex1)

Step 1: Convert temperature.

$$T = 15^{\circ}\mathrm{C} + 273\,\mathrm{K} = 288\,\mathrm{K}$$

Step 2: Find wave speed.

$$v_w = 331\sqrt{\frac{288}{273}}$$

$$v_w \approx 340\,\frac{\mathrm{m}}{\mathrm{s}}$$

Step 3: Find wavelength.

$$\lambda = \frac{v_w}{f}$$

$$\lambda = \frac{340\,\frac{\mathrm{m}}{\mathrm{s}}}{440\,\mathrm{Hz}}$$

$$\boxed{\lambda = 0.773\,\mathrm{m}}$$

Check Your Understanding

A sound wave has a frequency of $256\,\mathrm{Hz}$ traveling in air at $20°\mathrm{C}$ $(v_w \approx 343\,\mathrm{m/s})$.

The same wave then enters a steel rod $(v_w = 5960\,\mathrm{m/s})$. What happens to (a) the frequency and (b) the wavelength when it enters the steel?

Come to lecture to see the answer.

Human Hearing Frequency Range

Audible Frequency Range

- **Human hearing**: approximately 20 Hz to 20.000 Hz (20 kHz).
- **Infrasound**: below 20 Hz (inaudible; sometimes felt as vibrations).
- **Ultrasound**: above 20 kHz (inaudible to humans).

At 20°C, audible wavelengths range from ~17 m (at 20 Hz) to ~1.7 cm (at 20 000 Hz).

The range narrows with age; high-frequency sensitivity is lost first.

Animal Hearing Ranges

- Dogs: up to ~65 kHz
- Bats: up to ~200 kHz
- Dolphins: up to ~150 kHz
- Elephants: down to ~14 Hz (infrasound)

Check Your Understanding

The speed of sound in air at 30°C is approximately 349 m/s.

(a) What is the wavelength of a 20 Hz sound (low end of human hearing)?

(b) What is the wavelength of a 20,000 Hz sound (high end of human hearing)?

Come to lecture to see the answer.

Part II

Sound Intensity and the Doppler Effect

Outline of Part II: Sound Intensity and the Doppler Effect

Sound Intensity and Sound Level

Doppler Effect and Sonic Booms

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Sound Intensity and Sound Level

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Intensity of Sound

Definition: Intensity I —
SI Unit: watt per square meter (W/m^2)

Intensity is the power per area carried by a wave:

$$I = \frac{P}{A}$$

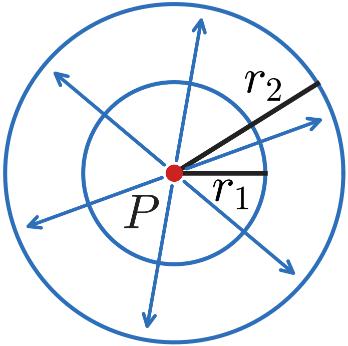
where P is the power (W) and A is the area over which it is distributed (m^2).

For a **point source** radiating in all directions:

$$A = 4\pi r^2 \;\Rightarrow\; I = \frac{P}{4\pi r^2}$$

$\therefore I \propto \frac{1}{r^2}$: double the distance, quarter the intensity.

$$A = 4\pi r^2$$



$$I = \frac{P}{4\pi r^2}$$

Intensity of Sound

Intensity from Pressure Amplitude

Sound intensity is also related to the pressure amplitude Δp:

$$I = \frac{(\Delta p)^2}{2\rho v_w}$$

where Δp is the pressure amplitude (Pa), ρ is the medium density ($\mathrm{kg/m^3}$).

$I \propto (\Delta p)^2$: doubling pressure amplitude *quadruples* intensity.

The Decibel Scale

Sound Intensity Level β — SI Unit: decibel (dB)

$$\beta\,(\mathrm{dB}) = 10\log_{10}\left(\frac{I}{I_0}\right)$$

where $I_0 = 1 \times 10^{-12}\,\frac{\mathrm{W}}{\mathrm{m}^2}$ is the **threshold of hearing** (reference intensity at 1000 Hz).

Why Use Decibels?

- It's natural $\because$ human hearing responds logarithmically to intensity.
- Audible intensities span a factor of 10^{12} (from 10^{-12} to $\sim 1\,\mathrm{W/m^2}$).
- Decibels compress this range into a convenient 0–120 dB scale.

Key Logarithm Rules

- $+10\,\mathrm{dB} \Rightarrow$ intensity $\times 10$
- $+3\,\mathrm{dB} \Rightarrow$ intensity $\times 2$
- $+20\,\mathrm{dB} \Rightarrow$ intensity $\times 100$
- $+30\,\mathrm{dB} \Rightarrow$ intensity $\times 1000$

Sound Levels in Everyday Life

Sound	β **(dB)**	I **(W/m^2)**
Threshold of hearing	0	10^{-12}
Rustling leaves	10	10^{-11}
Whisper	20	10^{-10}
Normal conversation	60	10^{-6}
Busy traffic	70	10^{-5}
Lawn mower	90	10^{-3}
Loud rock concert	120	1
Jet engine (30 m away)	140	100

Sustained exposure above 85 dB causes progressive, permanent hearing damage.

Example: Calculating Sound Intensity Level

Calculating Sound Intensity Levels: Sound Waves (cp2e_ch17_ex2)

Find β for a sound with $I = 2.00 \times 10^{-5}\,\frac{\mathrm{W}}{\mathrm{m}^2}$.

Given:

$$I = 2.00 \times 10^{-5}\,\frac{\mathrm{W}}{\mathrm{m}^2}$$

$$I_0 = 1.00 \times 10^{-12}\,\frac{\mathrm{W}}{\mathrm{m}^2}$$

$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$$

$$\beta = 10\log_{10}\left(\frac{2.00 \times 10^{-5}}{1.00 \times 10^{-12}}\right)$$

$$\beta = 10\log_{10}\left(2.00 \times 10^{7}\right)$$

$$\beta = 10\,(7.301)$$

$$\boxed{\beta = 73.0\,\mathrm{dB}}$$

Check Your Understanding

A community experiences background noise at $70\,\text{dB}$. A new highway would raise the level to $100\,\text{dB}$. How many times more intense is $100\,\text{dB}$ than $70\,\text{dB}$? Should the community be concerned?

Come to lecture to see the answer.

Check Your Understanding

One person talking produces 60 dB. What level do two people talking simultaneously produce?

Come to lecture to see the answer.

Check Your Understanding

The intensity from a point source drops to $\frac{1}{4}$ of its original value when you move farther away.

By how many decibels does the sound level change?

Come to lecture to see the answer.

Outline of Part II: Sound Intensity and the Doppler Effect

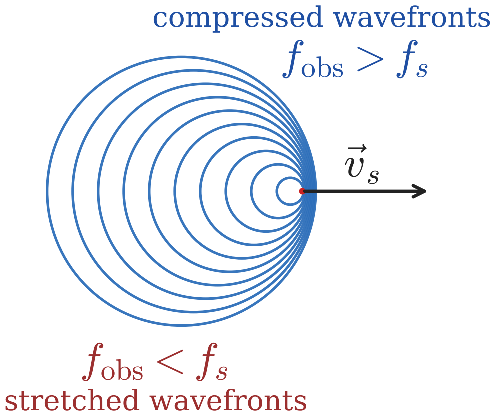
Sound Intensity and Sound Level

Doppler Effect and Sonic Booms

The Doppler Effect

The **Doppler effect** is the change in observed frequency of a sound (or any wave) caused by **relative motion** between the source and the observer.

- Ambulance siren: pitch is higher as it approaches, lower as it recedes.
- Race car: engine note drops as it passes.
- Police radar: measures vehicle speed from Doppler shift of reflected radio waves.
- Astronomy: redshift reveals that galaxies are receding from us.



Source approaching: $f_{\text{obs}} > f_s$

Source receding: $f_{\text{obs}} < f_s$

The Doppler Effect

Frequency of a Moving Source heard by a Stationary Observer

$$f_{\text{obs}} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$$

where f_s is the source frequency, v_w is the speed of sound, and v_s is the speed of the source.

☠ Check the Sign Convention!

- **Source approaching** observer:
  use minus $(-)$ in the denominator $\Rightarrow$ denominator smaller $\Rightarrow f_{\text{obs}} > f_s$.
- **Source receding** from observer:
  use plus $(+)$ in the denominator $\Rightarrow$ denominator larger $\Rightarrow f_{\text{obs}} < f_s$.

The speed of sound v_w in the medium is constant; relative motion changes the spacing between wave crests as perceived by the observer.

Example: Train Horn

Calculate Doppler Shift: A Train Horn (cp2e_ch17_ex4)

A train horn sounds at $f_s = 150\,\text{Hz}$. The train moves at $v_s = 35.0\,\text{m/s}$ in air where $v_w = 340\,\text{m/s}$. Find f_obs (a) as the train approaches and (b) as it recedes.

(a)

$$f_\text{obs} = f_s\left(\frac{v_w}{v_w - v_s}\right) = 150\left(\frac{340}{340 - 35}\right) = 150\left(\frac{340}{305}\right)$$

$$\boxed{f_\text{obs} = 167\,\text{Hz}}$$

(b)

$$f_\text{obs} = f_s\left(\frac{v_w}{v_w + v_s}\right) = 150\left(\frac{340}{340 + 35}\right) = 150\left(\frac{340}{375}\right)$$

$$\boxed{f_\text{obs} = 136\,\text{Hz}}$$

Check Your Understanding

Is the Doppler shift a physical change in frequency or merely a perceptual effect? How would you demonstrate this?

Come to lecture to see the answer.

Check Your Understanding

A police siren emits 800 Hz. The police car approaches you at $30.0\,\mathrm{m/s}$ with $v_w = 340\,\mathrm{m/s}$. What frequency do you observe?

Come to lecture to see the answer.

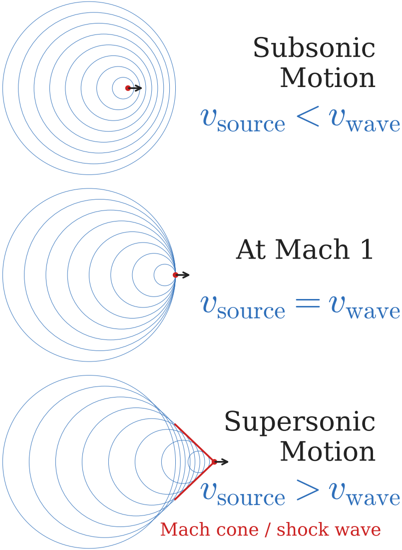
Sonic Booms

What Causes a Sonic Boom?

- When $v_s > v_w$, the source outruns its own sound waves.
- Waves pile up into a **shock wave** forming a cone (Mach cone) trailing behind the object.
- Two booms occur: one from the nose and one from the tail.
- The Mach cone sweeps along the ground continuously — not a one-time event.

$$\text{Mach number} = \frac{v_s}{v_w}$$

Sonic boom requires Mach number > 1.



Check Your Understanding

A supersonic jet passes directly overhead. You hear the sonic boom a few seconds after you can see that the jet has already traveled well past you.

Why is the plane already far past you when you hear the boom?

Come to lecture to see the answer.

Check Your Understanding

A supersonic jet passes directly overhead. You hear the sonic boom a few seconds after you can see that the jet has already traveled well past you.

Do you see the jet before or after you hear the boom? Why?

Come to lecture to see the answer.

Part III

Standing Waves in Air Columns

Outline of Part III: Standing Waves in Air Columns

Resonance and Standing Waves

Outline of Part III: Standing Waves in Air Columns

Resonance and Standing Waves

Resonance in Air Columns

Definition: Resonance

Resonance occurs when a system is driven at its **natural frequency**, allowing energy to build up and producing a standing wave with maximum amplitude.

Nodes and Antinodes

- **Node**: point of *zero* air displacement. Occurs at a **closed** end.
- **Antinode**: point of *maximum* air displacement. Occurs at an **open** end.
- A standing wave consists of alternating nodes and antinodes.

Boundary Conditions

Closed end $\Rightarrow$ **Node**
(wall prevents air displacement)

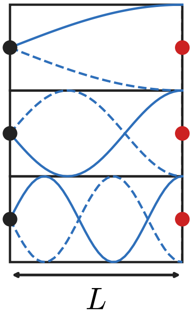
Open end $\Rightarrow$ **Antinode**
(air is free to move)

Tube Closed at One End

Resonant Frequencies: One End Closed $(n = 1, 3, 5, \ldots)$

$$f_n = n\frac{v_w}{4L}, \qquad n = 1,\ 3,\ 5,\ 7,\ \ldots$$

where L is the tube length and v_w is the speed of sound.



$n = 1$ $\quad f_1 = \dfrac{1v_w}{4L}$ $\quad L = \dfrac{1\lambda_n}{4}$

$n = 3$ $\quad f_3 = \dfrac{3v_w}{4L}$ $\quad L = \dfrac{3\lambda_n}{4}$

$n = 5$ $\quad f_5 = \dfrac{5v_w}{4L}$ $\quad L = \dfrac{5\lambda_n}{4}$

Why only odd harmonics?

The boundary conditions (node at closed end, antinode at open end) require an *odd* number of quarter-wavelengths to fit in the tube.

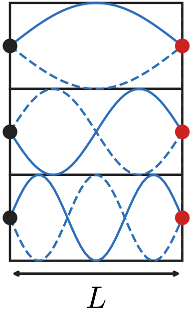
- Fundamental $(n = 1)$: $f_1 = \dfrac{v_w}{4L}, \quad \lambda_1 = 4L$
- 1st overtone $(n = 3)$: $f_3 = \dfrac{3v_w}{4L}, \quad \lambda_3 = \dfrac{4L}{3}$
- 2nd overtone $(n = 5)$: $f_5 = \dfrac{5v_w}{4L}, \quad \lambda_5 = \dfrac{4L}{5}$

Tube Open at Both Ends

Resonant Frequencies: Both Ends Open ($n = 1, 2, 3, \ldots$)

$$f_n = n\frac{v_w}{2L}, \qquad n = 1,\ 2,\ 3,\ 4,\ \ldots$$

where L is the tube length and v_w is the speed of sound.



$n = 1$ $\quad f_1 = \frac{1v_w}{2L}$ $\quad \lambda_1 = \frac{2L}{1}$

$n = 2$ $\quad f_2 = \frac{2v_w}{2L}$ $\quad \lambda_2 = \frac{2L}{2}$

$n = 3$ $\quad f_3 = \frac{3v_w}{2L}$ $\quad \lambda_3 = \frac{2L}{3}$

Why all harmonics?

Antinodes at *both* ends require an integer number of half-wavelengths in the tube. All positive integers satisfy this.

- Fundamental ($n = 1$): $f_1 = \frac{v_w}{2L}$, $\quad \lambda_1 = 2L$
- 2nd harmonic ($n = 2$): $f_2 = \frac{v_w}{L}$, $\quad \lambda_2 = L$
- 3rd harmonic ($n = 3$): $f_3 = \frac{3v_w}{2L}$, $\quad \lambda_3 = \frac{2L}{3}$

Closed vs. Open Tubes: Comparison

Closed Tube ($n = 1, 3, 5, \ldots$)

- Fundamental: $f_1 = \dfrac{v_w}{4L}$
- Lacks even harmonics
  $\implies$ more "hollow" sound
- Musical example: clarinet

Open Tube ($n = 1, 2, 3, \ldots$)

- Fundamental: $f_1 = \dfrac{v_w}{2L}$
- All harmonics present
  $\implies$ brighter, richer timbre
- Musical example: flute

For the *same length* L: $f_1(\text{open}) = 2 \times f_1(\text{closed})$

$\therefore$ the closed tube produces the lower fundamental frequency.

Harmonics vs. Overtones

Term	**Open tube**	**Closed tube**
1st harmonic (fundamental)	f_1	f_1
1st overtone	$2f_1$ (2nd harmonic)	$3f_1$ (3rd harmonic)
2nd overtone	$3f_1$ (3rd harmonic)	$5f_1$ (5th harmonic)
3rd overtone	$4f_1$ (4th harmonic)	$7f_1$ (7th harmonic)

- **Harmonics**: all resonant frequencies, including the fundamental.
- **Overtones**: only frequencies *above* the fundamental (excludes the fundamental).
- These terms are **not** interchangeable!

Example: Finding Tube Length from Resonance

A tube closed at one end resonates at its fundamental frequency of 128 Hz at 20°C ($v_w = 343$ m/s). Find the tube length.

Fundamental of a closed tube $(n = 1)$:

$$f_1 = \frac{v_w}{4L}$$

Solving for L:

$$L = \frac{v_w}{4f_1}$$

$$L = \frac{v_w}{4f_1}$$

$$L = \frac{343\,\frac{\mathrm{m}}{\mathrm{s}}}{4 \times 128\,\mathrm{Hz}}$$

$$L = \frac{343}{512}$$

$$\boxed{L = 0.670\,\mathrm{m}}$$

Check Your Understanding

(a) Which produces a lower fundamental frequency for the same tube length: a tube open at both ends, or a tube closed at one end? Explain.

(b) A flute (open tube) is 0.600 m long. What is its fundamental frequency at 20°C ($v_w = 343\,\mathrm{m/s}$)?

Come to lecture to see the answer.

Check Your Understanding

A guitar string vibrates at 440 Hz. The large wooden body of the guitar also vibrates when the string is plucked.

Why does a guitar produce much louder sound than just the bare string vibrating in open air?

Come to lecture to see the answer.

Part IV

Hearing and Ultrasound

Outline of Part IV: Hearing and Ultrasound

Outline of Part IV: Hearing and Ultrasound

The Physics of Hearing

Ultrasound

The Anatomy of the Ear

(Figure: cross-section of the human ear)

Three-Part Ear Structure

Outer Ear

- Pinna collects and funnels sound.
- Ear canal conducts sound to the eardrum.

Middle Ear

- Eardrum (tympanum) vibrates with incoming pressure waves.
- Three ossicles (malleus, incus, stapes) amplify pressure $\sim 40\times$ and transmit to inner ear's oval window.

Inner Ear

- Cochlea: fluid-filled spiral structure.

Pitch, Loudness, and Timbre

Pitch

- Perception of **frequency**.
- High $f \Rightarrow$ high pitch.
- Perceived logarithmically.
- Can detect $\sim 0.3\%$ frequency differences.

Loudness

- Perception of **intensity**.
- Measured in **phons**.
- *Frequency-dependent*: equal dB at different frequencies does not sound equally loud.

Timbre

- Tone quality.
- Determined by the **harmonic content**.
- Distinguishes instruments at the same pitch and loudness.

A violin and a flute playing the same note at the same volume are distinguished by **timbre** — the different mix of harmonics each instrument produces.

Equal-Loudness Curves

(Figure: Fletcher-Munson equal-loudness curves)

- Ear is **most sensitive** at 2,000–5,000 Hz.
- At low frequencies, *much higher* intensity is needed to sound equally loud.
- A 60 dB bass note sounds quieter than a 60 dB treble note.
- This is why bass-heavy music sounds "thin" at low volume but full at high volume.

Hearing Loss

Presbycusis (Age-Related)

- Progressive loss of hearing with age.
- Disproportionately severe at **high frequencies** (above 3,000 Hz).
- Caused by stiffening of cochlear structures and loss of hair cells.
- Not reversible.

Noise-Induced Hearing Loss

- Caused by prolonged exposure to sound > 85 dB.
- Characteristic "notch" in sensitivity near 4,000 Hz.
- **Permanent**: cochlear hair cells do not regenerate.
- **Preventable** with earplugs or earmuffs.

Wear hearing protection around loud machinery, concerts, or firearms.
Hearing loss is cumulative and irreversible.

Check Your Understanding

(a) Distinguish between **pitch**, **loudness**, and **timbre**. What physical property of sound does each correspond to?

(b) A piano and a guitar both play the note A (440 Hz) at the same intensity. Why do we perceive them as sounding different?

Come to lecture to see the answer.

Outline of Part IV: Hearing and Ultrasound

What is Ultrasound?

Definition: Ultrasound

Ultrasound is sound with frequency **above** 20.000 Hz (20 kHz), beyond the upper limit of normal human hearing.

Applications of Ultrasound

- **Medical imaging**: fetal scans, echocardiography, abdominal organs.
- **Doppler blood-flow**: measures blood velocity from frequency shift of echoes.
- **Therapeutic**: tissue heating, pain relief, wound healing.
- **Surgical**: kidney stone fragmentation (lithotripsy), tumor ablation.
- **Industrial**: non-destructive material

Resolution Limit:

We can never resolve details significantly *smaller* than the wavelength λ.

$\therefore$ Higher $f \Rightarrow$ shorter $\lambda \Rightarrow$ finer resolution.

Trade-off: higher f also means *less* penetration (more absorption).

Acoustic Impedance

Definition: Acoustic Impedance Z — SI Unit: kg/(m^2·s)

Each medium has an **acoustic impedance** Z defined as:

$$Z = \rho\, v_w$$

where ρ is the medium density (kg/m^3) and v_w is the speed of sound in that medium (m/s).

- When sound crosses from one medium to another, the **difference in impedance** determines how much reflects and how much transmits.
- Large mismatch $\Rightarrow$ strong reflection.
- Small mismatch $\Rightarrow$ mostly transmitted.

Medium	Z (kg/(m^{2}s))
Air	≈ 429
Fat tissue	$\approx 1.34 \times 10^6$
Muscle	$\approx 1.70 \times 10^6$
Bone	$\approx 7.80 \times 10^6$

Intensity Reflection Coefficient

Intensity Reflection Coefficient a (dimensionless, 0 to 1)

The fraction of sound intensity **reflected** at a boundary between two media:

$$a = \left(\frac{Z_2 - Z_1}{Z_1 + Z_2}\right)^2$$

where Z_1 and Z_2 are the acoustic impedances of the two media.

- $a = 0$: no reflection (impedances equal).
- $a = 1$: complete reflection (extreme mismatch).
- Fraction **transmitted** $= 1 - a$.

Why Ultrasound Gel is Necessary

Air: $Z \approx 429\,\mathrm{kg/(m^2 s)}$
Soft tissue: $Z \approx 1.5 \times 10^6$

This enormous mismatch gives $a \approx 0.9994$ ($\sim$99.9% reflected). Gel eliminates the air gap, allowing sound to enter tissue efficiently.

Example: Reflection at a Tissue Boundary

Find the intensity reflection coefficient at a fat-muscle boundary.
$Z_{\text{fat}} = 1.34 \times 10^6$ kg/(m^{2}s), $Z_{\text{muscle}} = 1.70 \times 10^6$ kg/(m^{2}s). (cp2e_ch17_ex7)

$$Z_1 = Z_{\text{fat}} = 1.34 \times 10^6$$
$$Z_2 = Z_{\text{muscle}} = 1.70 \times 10^6$$

$$a = \left(\frac{Z_2 - Z_1}{Z_1 + Z_2}\right)^2$$

$$a = \left(\frac{(1.70 - 1.34) \times 10^6}{(1.34 + 1.70) \times 10^6}\right)^2$$

$$a = \left(\frac{0.36}{3.04}\right)^2$$

$$a = (0.1184)^2$$

$$\boxed{a = 0.0140 \approx 1.40\%}$$

Only $\sim$1.4% of ultrasound is reflected; 98.6% penetrates into the muscle — good for imaging.

Ultrasound Intensity and Medical Safety

Application	Intensity (W/m^2)	Effect on Tissue
Diagnostic imaging	$\sim 10^{-2}$	None; safe
Therapeutic ultrasound	1–10	Gentle heating
Surgical/ablation	10^3–10^5	Cavitation; destruction

Diagnostic ultrasound uses intensities so low that no documented tissue damage occurs. Safety is well-established — the same physics, but at a tiny fraction of the power.

Cavitation: at high intensities, ultrasound creates vapor bubbles that collapse violently, destroying tissue.

Used intentionally in **lithotripsy** to fragment kidney stones without surgery.

Check Your Understanding

(a) Why is ultrasound imaging more difficult in patients with a higher body fat percentage?

(b) Do higher or lower ultrasound frequencies penetrate tissue better? What is the resolution trade-off?

Come to lecture to see the answer.

Check Your Understanding

Calculate the acoustic impedance of air given $\rho_{\text{air}} = 1.29\,\text{kg/m}^3$ and $v_w = 331\,\text{m/s}$.

Then calculate the reflection coefficient a at an air-tissue boundary where $Z_{\text{tissue}} = 1.50 \times 10^6\,\text{kg/(m}^2\text{s)}$. What does this confirm about the need for ultrasound gel?

Come to lecture to see the answer.

Check Your Understanding

Why is infrasound (below 20 Hz) useful for long-distance animal communication, such as among elephants?

What is one disadvantage of infrasound for communication?

Come to lecture to see the answer.